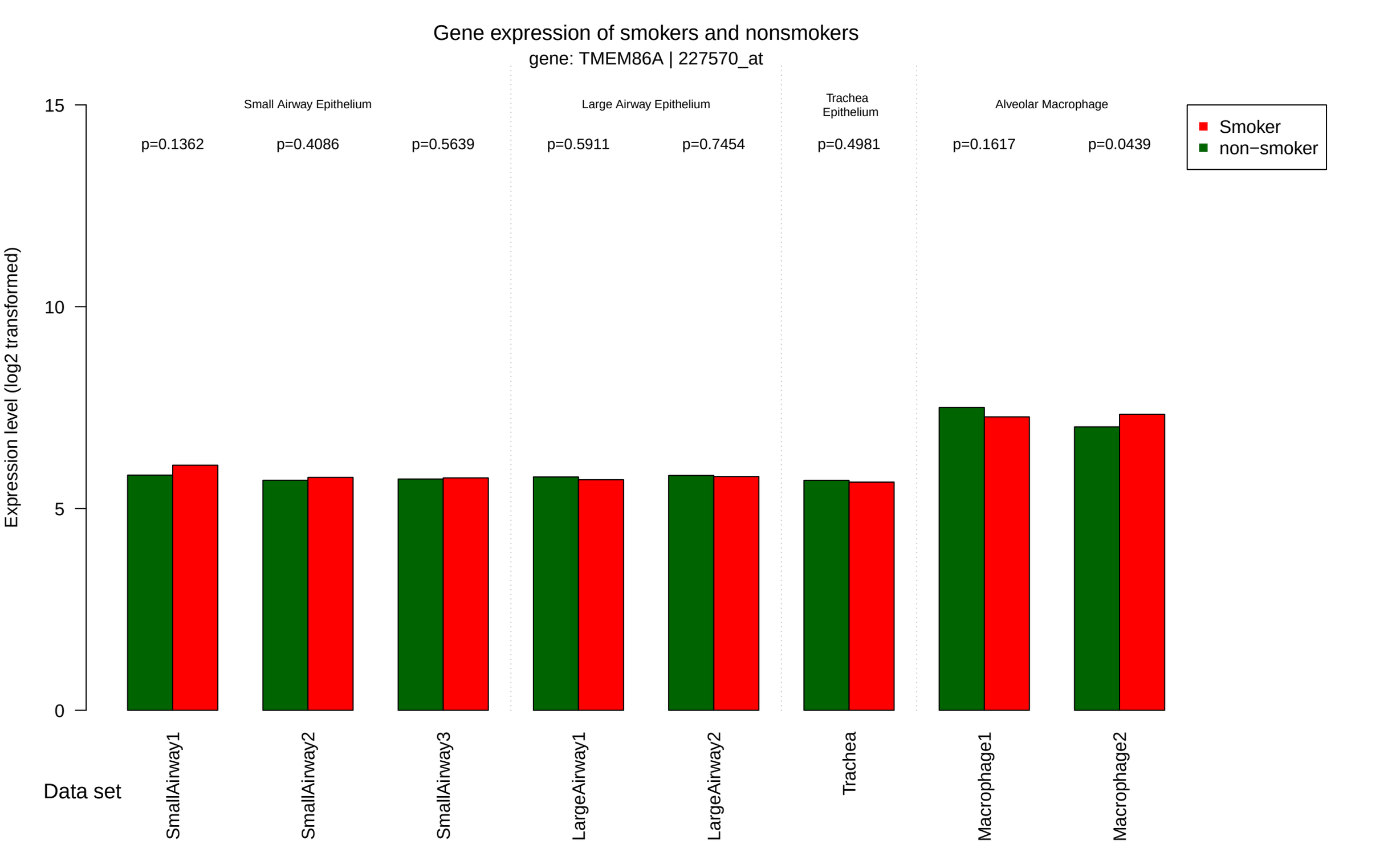
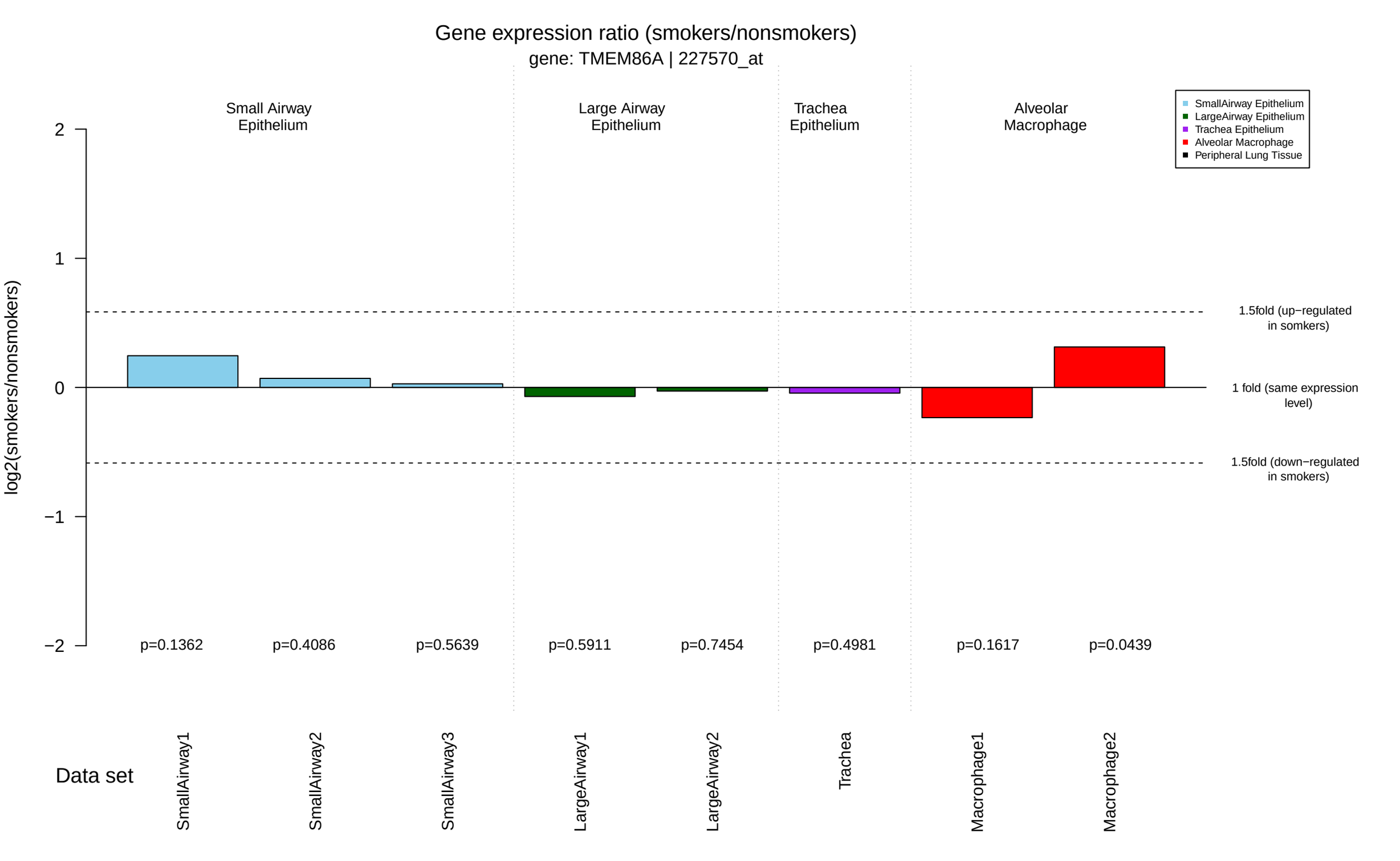
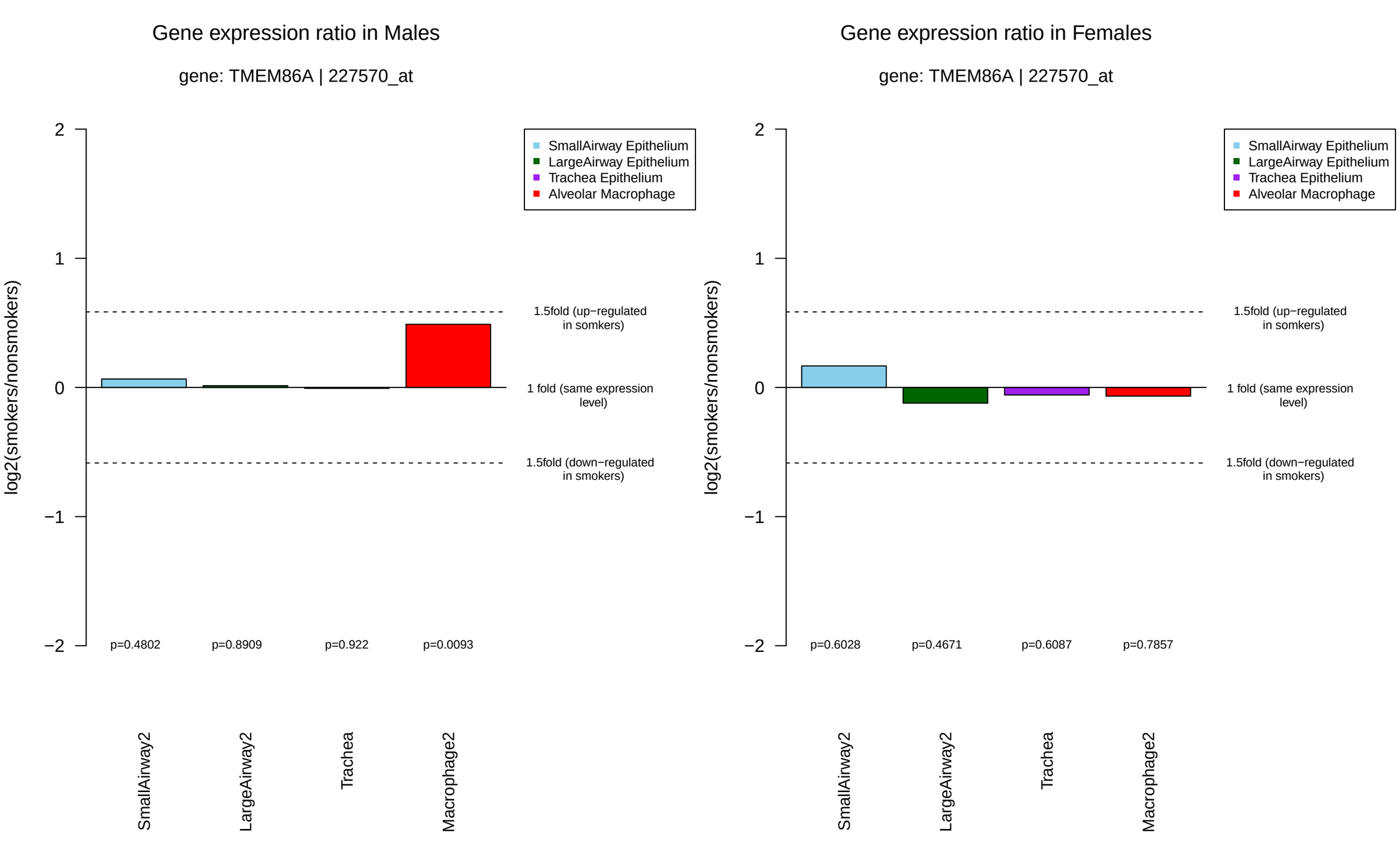
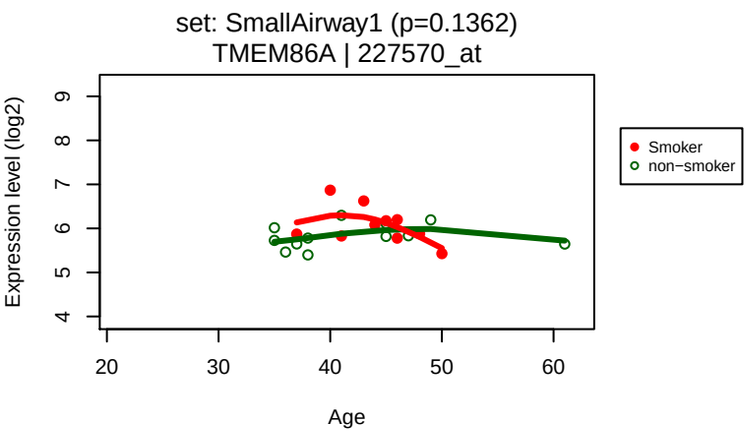
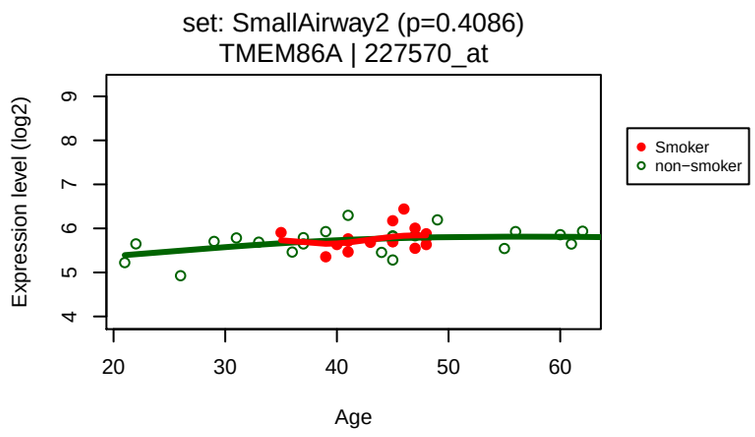
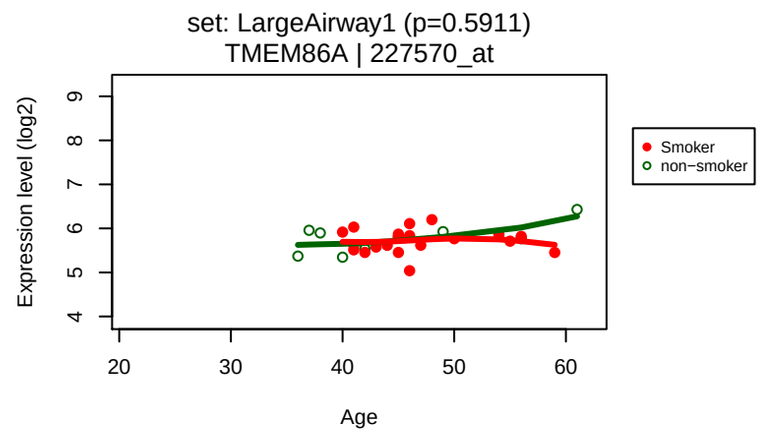

Gene expression of smokers and nonsmokers
gene: TMEM86A | 227570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1362
p=0.4086
p=0.5639
p=0.5911
p=0.7454
p=0.4981
p=0.1617
p=0.0439
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

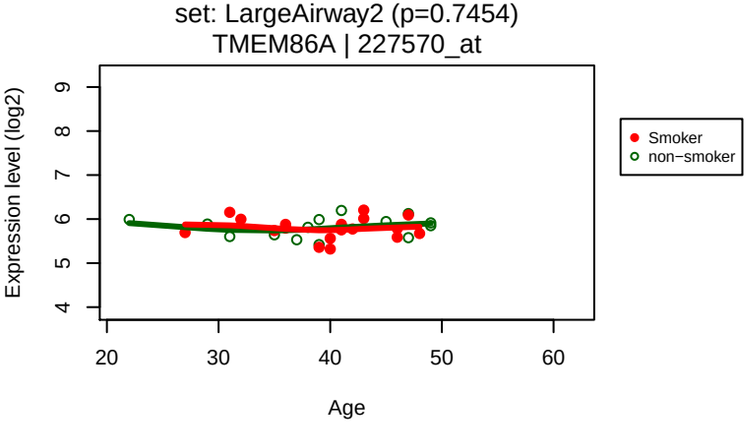

Gene expression ratio (smokers/nonsmokers)
gene: TMEM86A | 227570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1362
p=0.4086
p=0.5639
p=0.5911
p=0.7454
p=0.4981
p=0.1617
p=0.0439
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

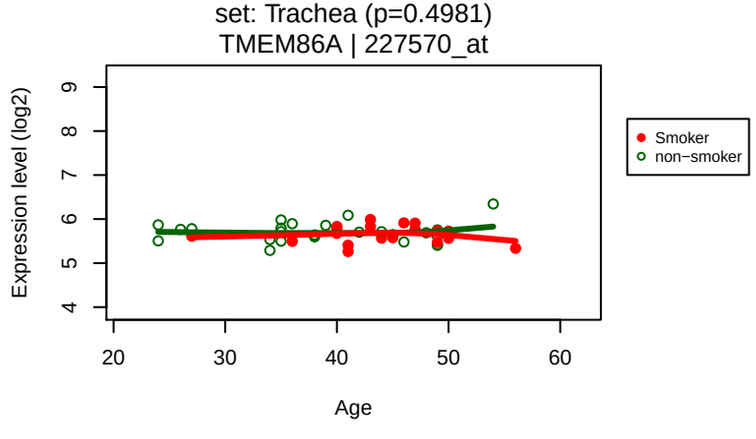

Gene expression ratio in Males
gene: TMEM86A | 227570_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4802
p=0.8909
p=0.922
p=0.0093
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM86A | 227570_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6028
p=0.4671
p=0.6087
p=0.7857
SmallAirway2
LargeAirway2
Trachea
Macrophage2

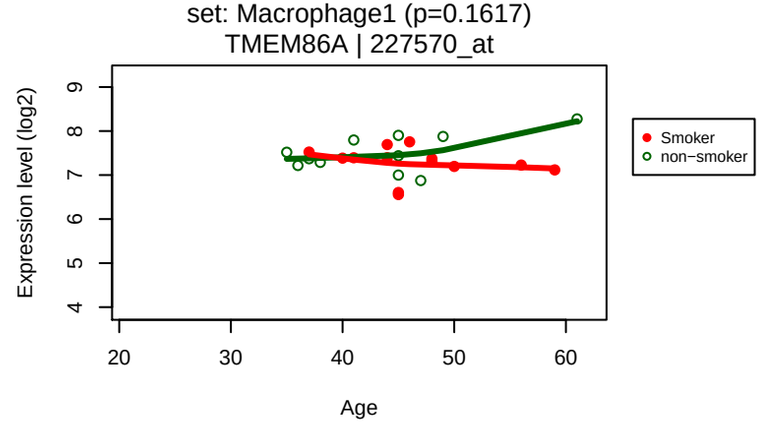

set: SmallAirway1 (p=0.1362)
TMEM86A | 227570_at
Expression level (log2)
Age
Smoker
non−smoker

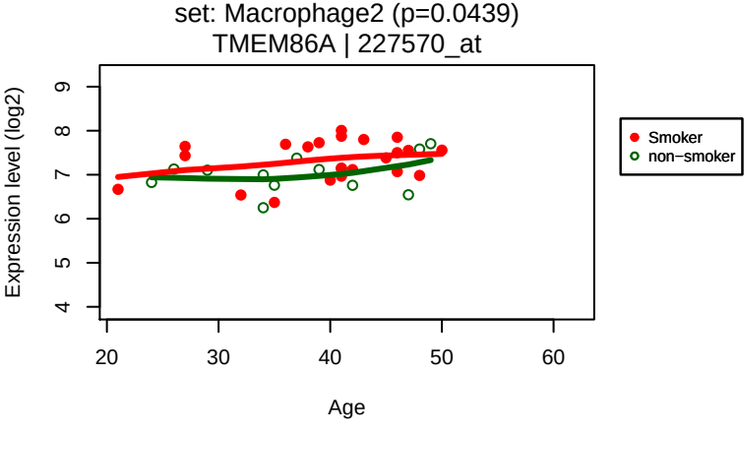

set: SmallAirway2 (p=0.4086)
TMEM86A | 227570_at
Expression level (log2)
Age
Smoker
non−smoker

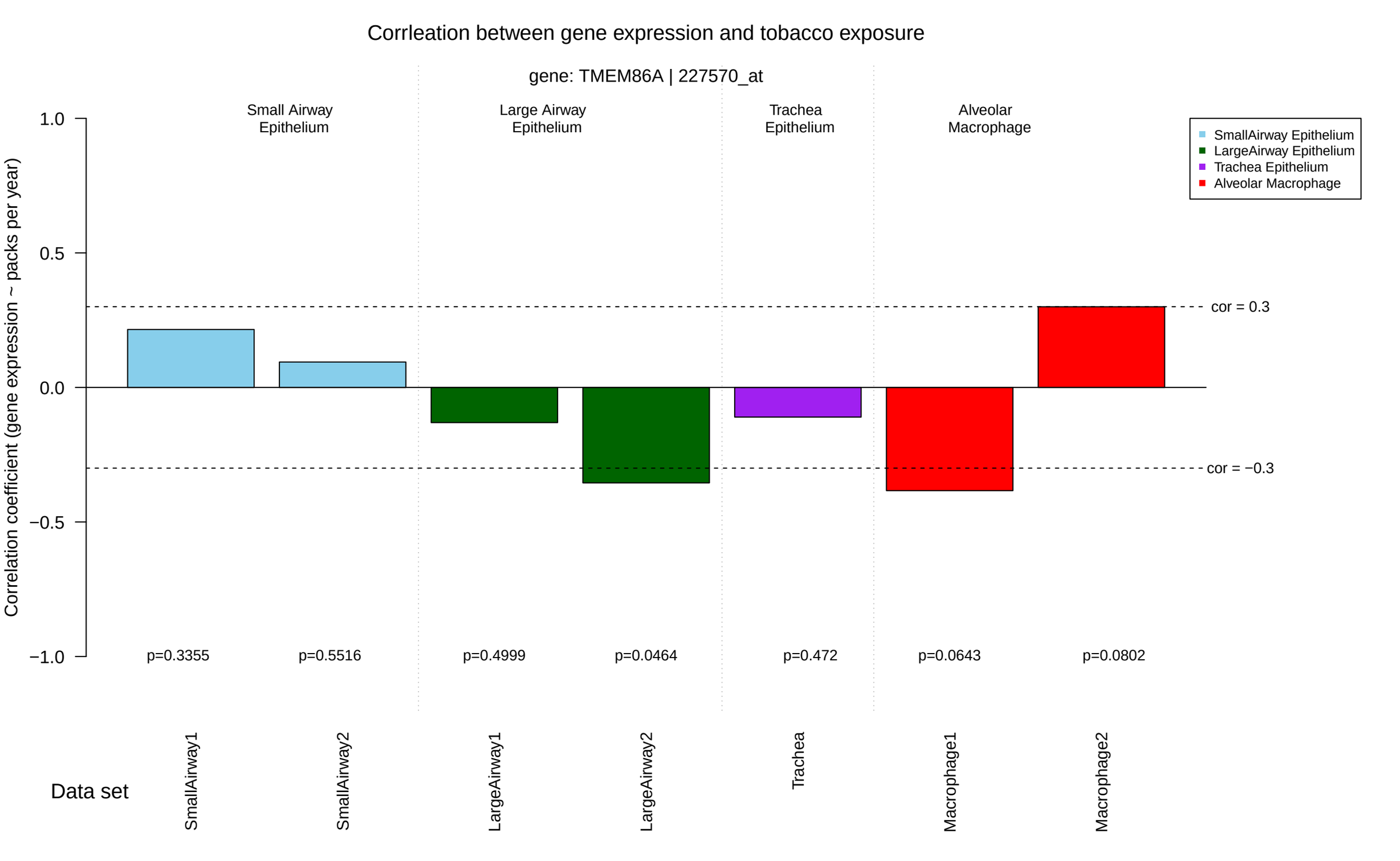

set: LargeAirway1 (p=0.5911)
TMEM86A | 227570_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7454)
TMEM86A | 227570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4981)
TMEM86A | 227570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1617)
TMEM86A | 227570_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0439)
TMEM86A | 227570_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM86A | 227570_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3355
p=0.5516
p=0.4999
p=0.0464
p=0.472
p=0.0643
p=0.0802
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2